

7-2b Using the Supply Curve to Measure Producer Surplus

Just as consumer surplus is closely related to the demand curve, producer surplus is closely related to the supply curve. To see how, let's continue with our example.

We begin by using the costs of the four painters to find the supply schedule for painting services. The table in [Figure 4](#) shows the supply schedule that corresponds to the costs in [Table 2](#). If the price is below \$500, none of the four painters is willing to do the job, so the quantity supplied is zero. If the price is between \$500 and \$600, only Andy is willing to do the job, so the quantity supplied is 1. If the price is between \$600 and \$800, Andy and Pablo are willing to do the job, so the quantity supplied is 2, and so on. Thus, the supply schedule is derived from the costs of the four painters.

Figure 4

The Supply Schedule and the Supply Curve

The table shows the supply schedule for the sellers (listed in [Table 2](#)) of painting services. The graph shows the corresponding supply curve. Note that the height of the supply curve reflects the sellers' costs.

The graph in [Figure 4](#) shows the supply curve that corresponds to this supply schedule. Note that the height of the supply curve is related to the sellers' costs. At any quantity, the price given by the supply curve shows the cost of the *marginal seller*, the seller who would leave the market first if the price were any lower. At a quantity of 4 houses, for instance, the supply curve has a height of \$900, the cost that Vincent (the marginal seller) incurs to provide his painting services. At a quantity of 3 houses, the supply curve has a height of \$800, the cost that Claude (who is now the marginal seller) incurs.

Because the supply curve reflects sellers' costs, we can use it to measure producer surplus. [Figure 5](#) uses the supply curve to compute producer surplus in our two examples. In panel (a), we assume that the price is \$600 (or slightly less). In this case, the quantity supplied is 1. Note that the area below the price and above the supply curve equals \$100. This amount is exactly the producer surplus we computed earlier for Andy.

Figure 5

Measuring Producer Surplus with the Supply Curve

In panel (a), the price of the good is \$600 and the producer surplus is \$100. In panel (b), the price of the good is \$800 and the producer surplus is \$500.

Panel (b) of [Figure 5](#) shows producer surplus at a price of \$800 (or slightly less). In this case, the area below the price and above the supply curve equals the total area of the two rectangles. This area equals \$500, the producer surplus we computed earlier for Pablo and Andy when two houses needed painting.

The lesson from this example applies to all supply curves: *The area below the price and above the supply curve measures the producer surplus in a market.* The logic is straightforward: The height of the supply curve measures sellers' costs, and the difference between the price and the cost of production is each seller's producer surplus. Thus, the total area is the sum of the producer surplus of all sellers.